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RESEARCH ARTICLE

Cyber hygiene knowledge, awareness, and behavioral practices of university students

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ABSTRACT

Personal data is valuable and vulnerable to individuals who have wrong intentions as any other possessions. This can be mitigated by good cyber hygiene habits. The main aim of this paper is to evaluate, analyze, and understand the level of cyber hygiene knowledge, awareness, and behavioral practices of university students and their mutual relations. We have conducted a survey study containing 30 cyber hygiene questions. Results show that students have acceptable cyber hygiene behavior, but their awareness is not satisfactory, and their knowledge is quite low. Also, the study shows the existence of some relations between gender and current education level and cyber hygiene knowledge, awareness, and behavior, as well as mutual interplay and relations between those cyber hygiene outcomes. The main contributions of this work are bidirectional: theoretical and practical.

KEYWORDS

awareness; behavior; cyber; hygiene; security

1. Introduction

The COVID-19 pandemic has seriously affected the European and international economy and society (Lallie et al., 2021). Governments have been expected to develop and implement novel economic plans (Padhan & Prabheesh, 2021). COVID-19 has also compelled organizations and individuals to accept new practices such as physical distancing, remote working, online learning, etc. However, while people around the globe have paid particular attention to economic and health threats caused by COVID-19, cyber attackers took advantage of this crises as well (Pranggono & Arabo, 2020).

Recent trends and cybersecurity statistics reveal a high growth in data breaches (Varonis, 2021). For example, in manufacturing industry, 92% attacks were financially motivated and 6% were motivated by espionage (Verizon, 2021). In addition, 82% of breaches in manufacturing industry were constituted by system intrusion, social engineering, and basic web application attacks (Verizon, 2021). Furthermore, ransomware was the fastest growing type of cyberattack whose frequency would continue to rise over next 5 years and reach every 2 seconds by 2031 (Cybersecurity Ventures, 2021). It is predicted that global cyberattacks costs will increase by 15% per year, reaching $10.5 trillion by 2025 (Cybersecurity Ventures, 2021). Nevertheless, total global data storage is predicted to exceed 200 zettabytes by 2025 (Cybersecurity Ventures, 2021). According to Cybersecurity Ventures, the imperative to protect digitized economy and society will initiate global cyber security spending to $1.75 trillion for 5-year period from 2021 to 2025. Furthermore, considering that our infrastructure (power, water, communication, etc.) relies on technology, it is clear why cybersecurity industry is planned to grow like no other (Jokić et al., 2021).

A rise in cyberattacks, especially phishing and malware, has been reported resulting in increasing number of infected devices (Baraković et al., 2016; Lallie et al., 2021). Not only organizations are being the aim of cyberattacks, but also individuals who use unfamiliar software for remote working or deal with online meeting requests (Hakak et al., 2020). Therefore, cybersecurity experts warn both organizations and individuals to ensure they do the right thing digitally. Just as we are focusing on hygiene

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with washing our hands, we need to focus on cyber hygiene not only during the COVID-19 pandemic, but also in the digital times to come. Additional motivation for that is the fact that we will rely on technology in coming years and our Quality of Life (QoL) will depend on digital world up to a great extent (Baraković et al., 2020), thereby making cybersecurity and cyber hygiene very important topics for all.

So far there have been several attempts to define cyber hygiene. In this regard, Maennel et al. (2018) summarized and compared cyber hygiene definitions provided in selected academic papers, government, and corporate documents. Since cyber hygiene has different interpretations in the academic literature, there is no common point of view whether it has technical or behavioral implications, or whether it is perceived at organizational or individual level. The non-academic literature covers definitions given by international organizations and industry initiatives, such as Center for Internet Security (CIS), National Institute of Standards and Technology (NIST), European Union Agency for Network and Information Security (ENISA), etc. All these definitions mainly focus on the organizational cyber hygiene considering technical aspects, whereas human aspects are covered to a limited extent (Esparza et al., 2020). In this paper, we adopted definition given by Vishwanath et al. (2019) according to who cyber hygiene is defined as "cybersecurity practices that online consumers should engage in to protect the safety and integrity of their personal information on their Internet enabled devices from being compromised in a cyber-attack."

In this regard, the main advantages of cyber hygiene are associated with maintenance and security. Maintenance is needed to decrease the risks of serious vulnerabilities, whereas security is necessary to predict and prevent hostile threats (e.g., viruses, malwares, spam, identity thief, phishing, etc.; Neigel et al., 2020). All hardware, software, and applications should be regularly maintained due to specific vulnerabilities that can lead to different problems including lost data, misplaced data, out-of-date software, older security software, etc. Since good cyber hygiene habits can help prevent these problems (Such et al., 2019), there is a need to evaluate the public knowledge, awareness, and behavioral practices in cybersecurity.

In this moment, there is no common standard or approach to cyber hygiene across European Union (EU) member states since each of them have their own programs which differ in level of maturity (ENISA, 2016). Bosnia and Herzegovina is a non-EU member state and has not published any cyber hygiene program so far. Therefore, it is unable to provide guidelines both for organizations and individuals to protect themselves from cyber threats in this country. However, given that Bosnia and Herzegovina aims to become EU member state (Baraković & Baraković Husić, 2015), eventually it will have to adopt and publish certain cyber hygiene program. To do so in a quality manner, it firstly must have research to determine the level of hygiene in cyber space of its citizens.

Therefore, the main objective of this study is to evaluate and analyze the level of cyber hygiene knowledge, awareness, and behavioral practices and their mutual relations of one group of citizens, i.e., students in Bosnia and Herzegovina. Since this type of study is the first one in Bosnia and Herzegovina (according to the authors' best knowledge), we have selected the university students for testing, because if we determine the knowledge, awareness, and behavior in this population who is educating to be future working force in cyber space, then we will be able to understand the gaps and improvement areas in broader population. Also, this paper discusses the good practices and the challenges they are facing. More specifically, the objectives of this study focus on:

- Understanding the cyber hygiene concepts and issues for supporting knowledge, awareness, and behavioral practices in cybersecurity in Bosnia and Herzegovina;
- Examining the existence of relations between gender and education level and cyber hygiene knowledge, awareness, and behavioral practices, as well as mutual interplay and relations between cyber hygiene outcomes in Bosnia and Herzegovina;

- Sharing good practices and proposing recommendations to appropriate stakeholders to promote the growth of cyber hygiene knowledge, awareness, and behavioral practices in Bosnia and Herzegovina.

This study provides insights for all cyber hygiene ecosystems. The relevant stakeholders may profit from the study's insights diversely, i.e., (i) research organizations, who perform research activities on cyber hygiene, (ii) public and private sector organizations, who need to support cyber hygiene practices, (iii) service providers, who are interested in user cyber hygiene services and technologies, and (iv) policy makers, who are concerned with defining cyber hygiene policies and recommendations.

The reminder of the paper is structured as follows. Section 2 provides related work on cyber hygiene, and its dimensions and outcomes. Section 3 presents the methodology adopted for performing the study. Section 4 gives the results analysis, while Section 5 provides discussion summarizing the study implications and limitations. Section 6 gives conclusion and recommendations.

2. Related work

Vishwanath et al. (2019) provided a conceptual definition and a multi-dimensional inventory called Cyber Hygiene Inventory (CHI) which is used to measure the cyber hygiene of internet users. The CHI contains items or questions that are categorized into five dimensions of user cyber hygiene, i.e., storage and device, authentication and credentials, Facebook and social media, e-mail and messaging, and transmission and browsing. This section categorizes related work on cyber hygiene according to aforementioned five dimensions, as shown in

Table 1. Related work on cyber hygiene dimensions and cybersecurity outcomes.

	Cyber Hygiene Dimensions						
Reference	Storage and Device Hygiene	Transmission and Browsing Hygiene	Facebook and Social Media Hygiene	Authentication and Credential Hygiene	E-mail and Messaging Hygiene	Objective	Methodology
Breitinger et al. (2020)	×					To test the level of behavior, awareness, and knowledge	Online survey study
Ovelgönne et al. (2017)	×					To test the level of behavior	Data-centered study
Gratian et al. (2018)	×	×		×	×	To test the level of behavior	Online survey study
Van Schaik et al. (2017)	×					To test the level of behavior	Online quantitative empirical study
Markelj and Bernik (2015)	×	×		×		To test the level of knowledge and awareness	Online survey study
Loukas and Patrikakis (2016)	×	×		×		To determine guidelines	Qualitative study
Imgraben et al. (2014)	×	×	×	×	×	To test the level of behavior, awareness, and knowledge	Paper/online survey study
Alsaleh et al. (2017)	×	×	×	×	×	To test the level of behavior and awareness	Quantitative and qualitative study (interviews)
Arachchilage and Love (2014)		×			×	To test the level of knowledge	Online questionnaire study
Debatin et al. (2009)			×			To test the level of awareness	Online survey study
Caputo et al. (2014)					×	To test the level of knowledge	Controlled experiment study
Cain et al. (2018)				×		To determine guidelines	Controlled experiment study
Florêncio et al. (2007)				×		To determine guidelines	Qualitative study
Dodge et al. (2007)					×	To test the level of awareness	Controlled experiment study

Table 1. This allows to summarize the related work in a wide and technology agnostic context. Additionally, it can be used to further discuss three aspects of cyber hygiene, i.e., knowledge, awareness, and behavior, and to provide a basis for strategic recommendations.

Storage and device hygiene refers to practices associated with the configuration settings of various devices, such as smartphones, desktop computers, etc. Examples of good hygiene involve running virus scan on any storage device, maintaining a backup of all important files, ensuring that device has latest operating system, and keeping the virus protection and software updated (Vishwanath et al., 2019). The use of security solutions derives from knowing the threats (Gratian et al., 2018). Students perceived identity theft as the highest risk followed by keylogger, cyber-bullying, and social engineering. Perceived risk was the lowest for cookie and e-mail harvesting (Van Schaik et al., 2017). Users are familiar with general threats, whereas they have poor knowledge of complex threats (Markelj & Bernik, 2015). The same is true for the use of security features. With fewer security features installed, smartphones are not secured as desktop computers (Breitinger et al., 2020. Young users secure their smartphones by setting lock screen protection but ignore other security practices increasing in that way the risk of malware attacks which software developers are the most vulnerable (Ovelgönne et al., 2017).

Transmission and browsing hygiene is associated with internet connections, since users are unaware of the risks related to connecting to an unknown or unsecure wireless-fidelity (Wi-Fi) network or leaving Wi-Fi turned on the whole time (Imgraben et al., 2014). Good practices include using virtual private network (VPN) while using a public Wi-Fi network, checking the quality of the secure socket layer (SSL) certificate, and checking the lock icon on the web browser (Vishwanath et al., 2019). These cyber hygiene practices should be enough to prevent most realistic cyber-physical attacks that can range from website advertisements to fraudulent Wi-Fi hotspots (Loukas & Patrikakis, 2016). Although users' protective behavior is affected by many factors, Alsaleh et al. (2017) showed that some practices are associated with each other and that young users behave more securely.

Facebook and social media hygiene are practices related to configuration and usage of social media. Some examples include assessing the authenticity of social media friend/information requests, knowing who you are connected to on social media, reassessing social/media friends/connection, managing privacy settings on social media platforms, ensuring the location information is not leaked in posts, etc. (Vishwanath et al., 2019). In support of these practices, Alani (2017) provided several cyber hygiene recommendations regarding rooting devices, installing applications, storing passwords, giving permissions, and malware threats. Debatin et al. (2009) showed that those recommendations contribute to the secure use of Facebook and social media and require alternations in user behavior. Modification of user behavior is demanding and requires embedded trainings which must consider perception of security support, preferential notification, information load, etc. (Caputo et al., 2014). Overall, He (2012) summarized the main practices used to prevent and reduce security risks, i.e., (i) developing a social media acceptable use and security practices, (ii) monitoring social media sites, (iii) monitoring employee's internet activity, (iv) providing education and training programs, (v) installing software updates, (vi) archiving social media content.

Authentication and credential hygiene describe activities related to use of secure browsers, changing of default username and passwords, managing how browser stores passwords, etc. (Vishwanath et al., 2019). In order to avoid over-the-shoulder attacks, Cain et al. (2018) studied different graphical authentication schemes and alphanumeric password. Similarly, Grawemeyer and Johnson (2011) studied the password usage and management in everyday life. It was found that lax security behavior included password reuse, lack of knowledge about the (in)secure passwords, sharing or writing down the passwords. Furthermore, Florêncio et al. (2007) found out that strong passwords did not protect the online users from password stealing attacks, such as keylogging and phishing. Weak passwords invited brute-force attack. However, increasing their strength did not address any real threat. It was better to increase the strength of the user identifications (ID's) rather than the passwords. Moreover, multi-factor authentication has appeared as an alternative practice to enhance security by requesting the user to provide more than one authentication factor contrary to only a password (Dasgupta et al., 2017). The most common instantiation of multi-factor authentication is based on two factors. A common assumption is that two-factor authentication has low usability, security, and deployability compared to authentication based only on passwords (Golla et al., 2021). Thus, authenticators and authentication factors need further investigation (Sinigaglia et al., 2020).

E-Mail and messaging hygiene includes practices such as creating a separate backup of all files on a backup disc, checking an incoming e-mail's header, checking sender's e-mail domain name, checking to see if e-mail requests have grammatical or typographical errors, etc. (Vishwanath et al., 2019). Some of those practices can help to avoid phishing attacks, so Dodge et al. (2007) organized phishing exercises served to give an insight into the level of awareness and concentrate to the awareness training. The combination of conceptual and procedural knowledge positively impacts self-efficiency which improved phishing threat avoidance behavior.

Summary of the related work indicates that the cyber hygiene practices are associated with each other, as shown in Table 1. Young users adopt some cyber hygiene practices but ignore others. Plenty of research activities have been conducted so far aimed to find the role of users in cybersecurity (ENISA, 2018). It is common view that user cyber hygiene can be measured by three user-oriented factors, i.e., knowledge, awareness, and behavioral practices. Knowledge and awareness refer to cognitive factors of knowledge and understanding, respectively, whereas behavioral practices are related to factor of use (Zwilling et al., 2020). Good cyber hygiene is based on these three factors: when educated users are aware of what must be done, they implement it when necessary.

Cybersecurity knowledge can be increased through cybersecurity training programs which would improve users' knowledge and translate it into effective cybersecurity behavior (Abawajy, 2014). For example, Arachchilage and Love (2014) studied the importance of conceptual and procedural knowledge in helping computer users to avoid phishing threats and found that it positively impacts self-efficiency which improved threat avoidance behavior. Furthermore, Zwilling et al. (2020) showed that cybersecurity knowledge is positively associated with cybersecurity awareness.

Cybersecurity awareness is a major factor in cybersecurity risk and can be described as low (e.g., neglecting security alerts), medium (e.g., negligence of improper technology operation), or high (e.g., knowing the cyber threats and their prevention; Zwilling et al., 2020). Shaw et al. (2009) noted lack of awareness of cyber risks in the context of application usage and information delivery on social networks and found out that attackers seek out the users with low level of cybersecurity awareness. To increase the cybersecurity awareness, there is a need to offer cybersecurity training programs. Dodge et al. (2007) showed that exercise raised cybersecurity awareness, making it difficult to separate out any increased awareness due to the training. Higher cybersecurity awareness is related to more cyber protection behaviors.

Cybersecurity behavior has become the focus of both academics and practitioners due to individual engagement generally and with cyber protection tools particularly (Zwilling et al., 2020). Ovelgönne et al. (2017) studied the problem of identifying cybersecurity behaviors which increase the risk of malware attacks on a host. The results indicate that machine features associated with human behavior were related to the number of malwares found on their host machines. On the other hand, correlation between human traits and cybersecurity behavior intention was studied by (Gratian et al., 2018). It was found that individual differences in demographic factors predicted from 5% to 23% of the change in cybersecurity behavior intentions. In addition, other traits, such as risk-taking preferences or decision-making style, were significantly correlated with good security behavior. Van Schaik et al. (2017) found that perceived control was a significant predictor of precautionary behavior.

Recent study conducted by Bong-Hyun and Ki-Chan (2017) has emphasized the importance of development cyber training programs in educational and academic institutions. Additionally, there are calls for more research to address low level of cybersecurity outcomes (i.e., knowledge, awareness, and behavior). Furthermore, it is expected that the advent of new paradigms such as internet of everything (IoE) will expand the security threats landscape and emphasize the need to determine the cybersecurity guidance (Loukas & Patrikakis, 2016). It is going to be about security protection by following strategies that have withstood the test of time (Cain et al., 2018) or intention to provoke discussion on the role of cyber hygiene practices (Florêncio et al., 2007). Therefore, the logical sequence of steps is to conduct an online survey with the aim of gathering information on cyber hygiene practices and then conducting controlled experimental study to test behavioral practices in a real-life scenario.

The risk-decreasing influence of good cyber hygiene practices motivates this study that seeks to investigate the users' knowledge, awareness, and behavioral practices across identified five dimensions of cyber hygiene in Bosnia and Herzegovina. Such investigation would contribute to the shift from technology- and process-oriented view to a user-oriented view of cyber hygiene and to the proposition of recommendation for cyber hygiene in Bosnia and Herzegovina.

3. Methodology

This section briefly explains the high-level methodology used to complete the survey, whose relevance was ensured in Section 2 by conducting the review of related work, and research questions.

3.1. Research questions

To achieve the abovementioned objectives, we have formulated the following research questions which refer to university student in Bosnia and Herzegovina:

- What is the level of cyber hygiene knowledge?
- What is the level of cyber hygiene awareness?
- What is the level of cyber hygiene behavioral practices?
- Is there a relation between gender and cyber hygiene knowledge, awareness, and behavioral practices?
- Is there a relation between education level and cyber hygiene knowledge, awareness, and behavioral practices?
- Is there a relation between cyber hygiene behavioral practices and awareness?
- Is there a relation between cyber hygiene behavioral practices and knowledge?
- Is there a relation between cyber hygiene knowledge and awareness?

3.2. Survey design

The survey was designed to collect demographic information and test the level of cyber hygiene knowledge, awareness, and behavioral practices of undergraduate and graduate students in Bosnia and Herzegovina, i.e., to provide answers to the previously posed research questions. Cyber hygiene is complex and diverse subject, but the survey questions were created to be generic and technology independent. In other words, all statements have been formulated in a simple manner to avoid the confusion. To develop the survey, the face validity was used. As such, author who is expert in cybersecurity education formulated the survey questions consulting the items from Olmstead and Smith (2017) and ; Terms of Use allow modification and usage of items.

The survey itself consisted of 30 questions which were categorized into five cyber hygiene dimensions (i.e., storage and device, authentication and credentials, Facebook and social media, e-mail and messaging, and transmission and browsing) and associated with three cyber hygiene outcomes (i.e., knowledge, awareness, behavior). Table 2 shows the mapping between the survey questions and cyber hygiene dimensions and outcomes. Also, Table 2 gives question types (short answer, checkbox, and multiple choice). Level of knowledge on each cyber hygiene dimension is tested through the following questions Q25 (D1 – storage and device), Q16-Q17, Q23, Q26 (D2 – transmission and browsing), Q18 (D3 – Facebook and social media), Q11, Q22, Q24 (D4 – authentication and credential), and Q19, Q21 (D5 – e-mail and messaging). Furthermore, level of awareness on specific cyber hygiene dimension is evaluated by questions Q7-Q8, Q20, Q27-Q30 (D1 – storage and device) and question Q15 (D2 – transmission and browsing). Finally, behavioral practices are assessed through questions Q3-Q6 (D1 – storage and device), Q13-Q14 (D2 –

Table 2. Mapping between the survey questions based on Olmstead and Smith (2017) and cyber hygiene dimensions and outcomes Vishwanath et al. (2019).

Cyber Hygiene Dimension	Questions	Question Type	Cyber Hygiene Outcome
D1 – storage and device	Q3	Checkbox	Behavior
D1 – storage and device	Q4	Checkbox	Behavior
D1 – storage and device	Q5	Checkbox	Behavior
D1 – storage and device	Q6	Checkbox	Behavior
D1 – storage and device	Q7	Multiple choice	Awareness
D1 – storage and device	Q8	Checkbox	Awareness
D1 – storage and device	Q20	Multiple choice	Awareness
D1 – storage and device	Q27	Multiple choice	Awareness
D1 – storage and device	Q28	Multiple choice	Awareness
D1 – storage and device	Q29	Multiple choice	Awareness
D1 – storage and device	Q30	Multiple choice	Awareness
D1 – storage and device	Q25	Multiple choice	Knowledge
D2 – transmission and browsing	Q16	Multiple choice	Knowledge
D2 – transmission and browsing	Q17	Multiple choice	Knowledge
D2 – transmission and browsing	Q23	Multiple choice	Knowledge
D2 – transmission and browsing	Q26	Multiple choice	Knowledge
D2 – transmission and browsing	Q18	Multiple choice	Knowledge
D2 – transmission and browsing	Q15	Multiple choice	Awareness
D2 – transmission and browsing	Q13	Checkbox	Behavior
D2 – transmission and browsing	Q14	Checkbox	Behavior
D3 – Facebook and social media	Q12	Checkbox	Behavior
D4 – authentication and credential	Q22	Multiple choice	Knowledge
D4 – authentication and credential	Q24	Multiple choice	Knowledge
D4 – authentication and credential	Q11	Multiple choice	Knowledge
D4 – authentication and credential	Q9	Multiple choice	Behavior
D4 – authentication and credential	Q10	Checkbox	Behavior
D5 – e-mail and messaging	Q19	Multiple choice	Knowledge
D5 – e-mail and messaging	Q21	Multiple choice	Knowledge
NA	Q1	Short answer	NA
NA	Q2	Short answer	NA

transmission and browsing), Q12 (D3 – Facebook and social media), and Q9-Q10 (D4 – authentication and credential).

In quantitative research, such as this one, one has to consider the validity of the used methods and measurements, i.e., in this case the questionnaire. Validity tells how accurately a method measures something. To determine if the research has validity, one has to consider construct, criterion, and content validity (Cronbach & Meehl, 1955). Given that we used no constructs in our questionnaire, there was no need to conduct the construct validity (convergent and discriminant). Also, since the research does not include criterions, there was no need to assess criterion validity (concurrent and

predictive). Finally, content validity has been addressed. The content validity assesses whether a test is representative of all aspects. There is no direct measure of content validity, and therefore it was tested by security and information and communication technology (ICT) experts who reviewed and rated the statements, removing the ones that were marked as irrelevant and accepting and adjusting the ones that were relevant. In addition, we have performed a pre-study survey with 25 participants to validate the study and questionnaire design.

3.3. Survey participants

In Bosnia and Herzegovina there is 74012 undergraduate and graduate students studying at universities (Agency for Statistics of Bosnia and Herzegovina, 2021). The determined sample size of the population of university students at confidence level of 95% and confidence interval of ±5% is 383. However, the survey included 509 participants meaning that the determined sample size was representative in terms of general population. A total of 88.14% (N = 450) of students were enrolled in the bachelor's degree (BSc) and 11.60% (N = 59) of students in the master's degree (MSc). The gender split was from 50.69% (N = 258) male and 49.31% (N = 251) female. The age range for participants was 18 to 28 years (M = 19.74 years; SD = 1.46 years). Demographic information is included in Table 3.

Table 3. Participant demographic information (N = 509).

Characteristics	Frequency	Proportion (%)
Sex		
Female	251	49.31%
Male	258	50.69%
Academic degree enrolment		
Bachelor's degree (BSc)	450	88.41%
Master's degree (MSc)	59	11.60%

3.4. Survey procedure

Participants were students recruited from the University of Sarajevo over the winter semester courses (first semester of BSc and MSc studies). They were introduced to the study goals, benefits, and procedures, and informed that the information they gave would be confidential. Then participants who were interested in taking part were asked to complete the paper-based survey within approximately 15–20 minutes during lecture breaks. Each participant took part in the survey on a fully free basis. The survey contained an information sheet and questionnaire survey. All information was gathered anonymously. Data collected by survey were inserted into Microsoft Excel and then analyzed by Statistical Package for the Social Sciences (SPSS) Statistics survey ("IBM SPSS," 2020) using descriptive statistics and Chi-square test where it was adequate.

4. Results

The results in this section are given in several tables, i.e., Tables 4–10, while Table 11 summarizes the resulting answers on research questions. Table 4 shows the summary of survey results through percentages. Tables 5–7 show the summaries of Chi-square test results related to relations of gender and current education level and cyber hygiene knowledge, awareness, and behavior organized by cyber hygiene dimensions (D1-D4, while D5 contains only 2 items so the results were reported in the text). Tables 8–10 cover results related to mutual interplay and relations between cyber hygiene knowledge, awareness, and behavioral practices obtained by conducting Chi-square test as well (behavior and awareness, behavior and knowledge, and awareness and knowledge).

Chi-square test is used to discover if there is a relationship between two categorical variables. Our variables are gender, current education level, and answers to survey questions. Therefore, the data satisfies the criteria for use of Chi-square test, i.e., all considered variables are categorical and consist of two or more independent groups. It is important to note that the relations between variables are considered statistically significant if p-value is less or equal 0.05 and they are marked in abovementioned tables.

According to Table 2, survey questions which correspond to behavior as cyber hygiene outcome and storage and device as cyber hygiene dimension are: Q3 – Q6. The results show that most users (regardless of being male or female, or BSc or MSc students) use desktop or laptop computers (96.3%) and smartphones (96.1%) to access the Internet (Table 4). Those computers are mainly accessed by the individuals exclusively (51.1%) or by their siblings as well (33%). However, it is interesting to note that the behavior of male and female students and BSc and MSc students in this context is similar, except when it comes to sharing their computer devices with parents where BSc students are more likely to do that (12% more in comparison to MSc students) and sharing with friends where MSc students are more likely to do that (4% more in comparison to BSc students; Table 5). On the other hand, most smartphones are exclusively used by the individuals (89.8%). However, among those nearly 10% of students who share their smartphone devices with others (parents, siblings, and friends), the female students dominate.

Further, the students mostly use their smartphones for social networks, Internet search, learning, games, shopping, financial management, and security management, respectively (Table 4). Male and female students proportionally use social networks and Internet search, while male students use it more in the context of shopping (17% more), financial management (15% more), games (20% more), and security management (20% more). Female students use smartphones for learning more than male colleagues (12% more; Table 5). These results show that the gender may have the impact on smartphone sharing behavior and application usage (Chi-square test p-value less than 0.05).

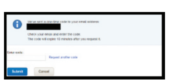
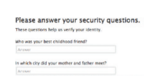
Storage and device dimension which results in cyber hygiene outcome – awareness is addressed by survey questions Q7, Q8, Q20, Q27-Q30. Namely, regarding the usage of security solutions such as antivirus or antimalware software on devices used for Internet, 56.9% of participants declared that they use them on all devices, 32.4% on computers, while only 5.3% on smartphones. 5.3% of the ones

Table 4. Summary of survey results.

	SURVEY QUESTIONS AND RESULTS							
Q1	How old are you?				Mean			
					19.73			
Q2	What gender are you?				Male	Female		
					50.6%	**49.4%**		
Q3	The devices I use to access the Internet are:							
	Desktop/laptop computer	Tablet	Smartphone	Smart wearables	Other			
	96.3%	**15.9%**	**96.1%**	**10%**	**2.6%**			
Q4	The computer I use can be accessed by							
	Me, exclusively	Me and my parents	Me and my siblings	Me and other extended family members	Me and my girlfriend/ boyfriend	Me and friends	All	
	51.1%	**24%**	**33%**	**3.5%**	**2.9%**	**2.9%**	**4.3%**	
Q5	The smartphone/ device I use can be accessed by							
	Me, exclusively	Me and my parents	Me and my siblings	Me and other extended family members	Me and my girlfriend/ boyfriend	Me and friends	All	
	89.8%	**3.7%**	**2.3%**	**0.2%**	**4.3%**	**3.1%**	**1%**	
Q6	The applications I have on my smartphone/ device are used for							
	Internet search	Social networks	Shopping	Financial management	Games	Security management	Learning	Other
	87.8%	**98.4%**	**40.1%**	**21.6%**	**57.3%**	**21.2%**	**72.5%**	**7.4%**
Q7	On which devices that you use are security solutions enabled (antivirus, antimalware, etc.)?							
	At all	On the computer	On a smartphone/device	Not at all	Other			
	56.9%	**32.4%**	**5.3%**	**5.3%**	**0.1%**			
Q8	If you don't use security solutions on your smartphone/device or computer, why don't you do it?							
	I hate it	There is no need	Because it takes up too much storage space	Too many options from which I don't know which one to choose	Too expensive	It takes too much time	Slows down the device	Other
	14.7%	**13.9%**	**6.8%**	**3.7%**	**2.2%**	**0.9%**	**10.4%**	**9%**
Q9	I create my passwords . . .							
	By combining letters, numbers and symbols and 8 characters	With at least 8 characters	From private information (date of birth, name, etc.)	All of the above	None of the above			
	50.8%	**7.7%**	**10%**	**20.4%**	**10.6%**			
Q10	How do you manage your passwords?							
	I store my passwords on a computer or write them down on paper	I use the same passwords for most occasions	I store the password on a browser (Mozilla, Chrome, etc.)	I do not hide my passwords – everyone knows them	I remember them	I store my passwords with the help of password manager		
	25.3%	**37.7%**	**6.2%**	**0.8%**	**53.6%**	**4.1%**		
Q11	Which of the following passwords is the most secure?							
	Most123	PoG#&78	lokum78	123456	Admin	I am not sure		
	4.7%	**84%**	**3.3%**	**1.6%**	**0.5%**	**5.9%**		
Q12	What personal information do I share through the devices I use to access the Internet? This includes social networks (Instagram, Facebook, etc.), e-mail (gmail.com, etc.), communication applications (Viber, WhatsApp, etc.), web search, etc.							
	Name and surname	Registration number	Images	Video	Documents	Opinions, feelings, statements, etc.		
	95.5%	**4.1%**	**90.2%**	**48.3%**	**15.9%**	**42.6%**		
	Home address	Relationship status	Phone number	Location	Bank card number	Plans	Other	
	16.5%	**15.5%**	**27.3%**	**31%**	**2.2%**	**5.1%**	**2.4%**	
Q13	Do you use public Wi-Fi networks?							
	I don't use public Wi-Fi networks	I use password-protected public Wi-Fi networks	I use public Wi-Fi networks that are familiar to me	I use public Wi-Fi networks that are available and free				
	6.3%	**49.1%**	**37.7%**	**60.3%**				

(*Continued*)

Table 4. (Continued).

SURVEY QUESTIONS AND RESULTS						
Q14	I use Wi-Fi network …					
	For social networks (including Viber, WhatsApp, etc.)	For e-mail communication	For online shopping		For banking transactions	For other applications
	96.3%	**63.6%**	**31.6%**		**17.7%**	**46%**
Q15	If a public Wi-Fi network (café, station, etc.) requires an access password, is it safe to use it for sensitive activities such as online or mobile banking?					
	Yes		No			Not sure
	10.8%		**48.3%**			**40.9%**
Q16	All Wi-Fi traffic is encrypted by default on all wireless routers.					
	No		Yes			Not sure
	16.3%		**24.4%**			**59.3%**
Q17	What does at the beginning of the URL mean compared to					
	That the website has high definition	That the entered information is encrypted	That it is the latest available version of the website	That the website cannot be accessed from certain computers	None of the above	Not sure
	13.9%	**24.4%**	**2.5%**	**3.5%**	**7.6%**	**48.1%**
Q18	"Private search" is a feature of many Internet browsers that allows users to access the website without any information (such as search history) that the browser can store. Do Internet Service Providers (ISPs) see their subscribers' online activities when they use the "private search" option?					
	Yes		No			Not sure
	48.7%		**12.2%**			**39.1%**
Q19	All e-mails are encrypted by default					
	Correct		Incorrect			Not sure
	49.3%		**7.5%**			**43.2%**
Q20	Do you check that the files you download or the devices you use are infected with viruses before you use them?					
	Yes, both	Files only	Devices only		No, because there is no need	No, because it takes too much time
	59%	**9.6%**	**5.6%**		**11%**	**14.8%**
Q21	Which of the following is an example of phishing?					
	When someone is sent an e-mail that contains a malicious link disguised as looking like an e-mail from a celebrity	Creating a fake website that looks approximately identical to the actual website in order to deceive the user and enter their information	When a text message is sent to someone containing a malicious link disguised as looking like a notification that a person has won a contest or lottery		All of the above	Not sure
	9%	**17%**	**4.5%**		**12.8%**	**56.7%**
Q22	Some websites and online services use a security process called two-factor authentication. Which of the following images is an example of two-factor authentication?					
					None of the above	Not sure
	15.3%	**22%**	**16.1%**	**10.6%**	**1.6%**	**34.4%**
Q23	A group of computers that are networked and used by hackers to steal information is called:					
	Botnet	Rootkit	DDoS		Operating system	Not sure
	16.2%	**10.7%**	**15%**		**3.4%**	**54.7%**
Q24	Criminals enter the computer and encrypt the user's personal files and data. The user is unable to access this data unless he pays criminals to decrypt the files. This is called:					
	Botnet	Ransomware	DDoS	Spam	None of the above	Not sure
	4.5%	**17.3%**	**8.2%**	**9.6%**	**1.8%**	**58.6%**
Q25	Turning off the Global Positioning System (GPS) feature on your smartphone makes it impossible to track your phone's location.					
	Correct		Incorrect			Not sure
	42.2%		**39.3%**			**18.5%**
Q26	What types of security risks are reduced by using virtual private networks (VPNs)?					
	Phishing attacks	Monitoring by website operators	Use of insecure Wi-Fi networks	Deanonymization by network providers	Writing keys	Not sure
	9.3%	**19.7%**	**10.1%**	**3.8%**	**2.7%**	**54.3%**

(*Continued*)

Table 4. (Continued).

SURVEY QUESTIONS AND RESULTS					
Q27	I back up my data on devices . . .				
	Once a month	Once in 3 months	Once in 6 months	Once a year	Never
	30.8%	**15.7%**	**11.5%**	**15.3%**	**26.7%**
Q28	Do you regularly update software on the smart devices you use?				
	Yes	No, because it is not necessary	No, because it takes too much time	No, because it doesn't matter to me	
	78.8%	**7.7%**	**5%**	**8.5%**	
Q29	The links I use most often to download software and applications on my computer are:				
	Checked and safe	Unverified and insecure	It depends		
	48.9%	**9.6%**	**41.5%**		
Q30	The links I use most often to download software and applications on my smartphone/ device are:				
	Checked and safe	Unverified and insecure	It depends		
	55%	**5.3%**	**39.7%**		

Abbreviations: DDoS, Distributed Denial of Service; URL, Uniform Resource Locator; Wi-Fi, Wireless Fidelity.

Table 5. Chi-square test results for storage and device dimension (D1) questions related to cyber hygiene knowledge, awareness, and behavioral practices.

	Gender				Current Education Level			
	X-value	p-value	% Male	%Female	X-value	p-value	% BSc	% MSc
Q4.1 (B)	1.214	0.271	53.5	48.6	1.150	0.284	49.9	55.8
Q4.2 (B)	0.058	0.809	24.4	23.5	**5.285**	**0.022**	**26.2**	**15.4**
Q4.3 (B)	3.666	0.056	29.1	37.1	0.096	0.757	33.3	31.7
Q4.4 (B)	0.177	0.674	3.9	3.2	0.163	0.687	3.7	2.9
Q4.5 (B)	0.100	0.752	2.7	3.2	1.582	0.208	2.5	4.8
Q4.6 (B)	0.100	0.752	2.7	3.2	**6.543**	**0.011**	**2.0**	**6.7**
Q4.7 (B)	0.252	0.616	3.9	4.8	0.072	0.789	4.4	3.8
Q5.1 (B)	**13.086**	**< 0.001**	**94.6**	**84.9**	2.522	0.112	90.9	85.6
Q5.2 (B)	**6.935**	**0.008**	**1.6**	**6.0**	0.262	0.609	4.0	2.9
Q5.3 (B)	**8.820**	**0.003**	**0.4**	**4.4**	**6.609**	**0.010**	**1.5**	**5.8**
Q5.4 (B)	0.975	0.323	0.4	0.0	0.257	0.612	0.2	0.0
Q5.5 (B)	**5.044**	**0.025**	**2.3**	**6.4**	3.590	0.058	3.5	7.7
Q5.6 (B)	1.149	0.284	2.3	4.0	0.212	0.645	3.0	3.8
Q5.7 (B)	1.903	0.168	0.4	1.6	4.863	0.127	0.5	2.9
Q6.1 (B)	3.034	0.082	90.3	85.3	0.804	0.370	87.2	90.4
Q6.2 (B)	**4.407**	**0.036**	**97.3**	**99.6**	0.315	0.575	98.3	99.0
Q6.3 (B)	**16.712**	**< 0.001**	**48.8**	**31.1**	2.009	0.156	38.5	46.2
Q6.4 (B)	**13.796**	**< 0.001**	**28.3**	**14.7**	0.437	0.509	22.2	19.2
Q6.5 (B)	**21.712**	**< 0.001**	**67.4**	**47.0**	0.663	0.416	58.3	53.8
Q6.6 (B)	**29.996**	**< 0.001**	**31.0**	**11.2**	2.661	0.103	22.7	15.4
Q6.7 (B)	**3.971**	**< 0.001**	**68.6**	**76.5**	0.348	0.555	73.1	70.2
Q6.8 (B)	0.181	0.671	7.0	8.0	3.971	0.066	8.6	2.9
Q7 (A)	8.176	0.085	-	-	4.089	0.394	-	-
Q20 (A)	**11.589**	**0.021**	**60.3**	**58.6**	**12.581**	**0.014**	**59.6**	**58.7**
			11.9	**7.2**			**9.8**	**8.7**
			4.0	**7.2**			**6.3**	**2.9**
			12.7	**8.8**			**12.0**	**5.8**
			11.1	**18.3**			**12.3**	**24.0**
Q25 (K)	**18.435**	**< 0.001**	**48.2**	**29.9**	0.350	0.840	38.7	40.4
Q27 (A)	1.554	0.817	-	-	3.759	0.440	-	-
Q28 (A)	1.275	0.100	-	-	3.458	0.484	-	-
Q29 (A)	1.270	0.530	-	-	2.138	0.343	-	-
Q30 (A)	0.120	0.942	-	-	2.437	0.296	-	-

Abbreviations: A, Awareness; B, Behavior; BSc, Bachelor's degree; K, Knowledge; MSc, Master's degree; Q, Question.

that do not use this software at all say that they do not use it because they hate it (14.7%), or there is no need to (13.9%), or it slows down the device (10.4%), while 9% of them stated other reasons for not using security software on their devices. There is no statistically significant difference between male and female students regarding this matter, as well as between BSc and MSc students (Table 5).

Furthermore, most of participants are aware of the dangers posed by downloaded files and used devices since 59% of them scan both, while 9.6% scan files only and 5.6% scan devices only. There are differences between male and female students regarding these activities: more male students scan files only, while more female students scan devices only. Also, there is more male students who consider that scanning the device or files is not needed (3.9% more in comparison to female students), while more female students do not perform these activities because it takes too much time (9.2% more in comparison to male students). When it comes to education level, the situation is similar: more BSc students think that scanning the device or files is not needed (6.2% more than MSc students), while more MSc students think that it takes too much time (11.7% more BSc students).

Results regarding data backups are concerning, since almost one third of participants never perform backups, while only one third of participants perform them once a month. The remaining third performs data backup in time frame from 3 months to one year. The same situation is with links used for software and application download on computers and smartphones. Namely, half of participants use checked and safe links (48.9% in case of computer and 55% in case of smartphones), while half of them use unverified and insecure or stated that it depends on the situation. Situations is better when it comes to software updates, since 78.8% of participants perform regular backups on smart devices they use. According to the results of Chi-square test given in Table 5, there is not statistically significant difference between male and female students and BSc and MSc students in this matter.

In terms of knowledge (as a cyber hygiene outcome) and storage and device hygiene (dimension), only 39.3% participants answered correctly to the question related to global positioning system (GPS) feature and tracking possibilities (Q25), while the remaining percent stated they were not sure (18.5%) or answered incorrectly (42.2%). 48.2% male students and 29.9% female students answered correctly, which indicates that gender is important in this context. There was no statistically significant association between education level and this item in cyber hygiene knowledge (Table 5).

Further on, we move to results in domain of transmission and browsing as cyber hygiene dimension (Q13-Q18, Q23, Q26). In terms of cyber hygiene outcome – behavior, Q13 and Q14 addressed the use of public Wi-Fi networks. Results show that 60.3% of participants use public Wi-Fi networks that are available and free, 49.1% use password protected public Wi-Fi networks, and 37.7% use Wi-Fi networks that are familiar to them. According to Table 6, there is no statistically significant association between gender and current education level and usage of Wi-Fi networks. Social networks are the prominent reason for Wi-Fi network usage (96.3%), which is followed by e-mail communication (63.3%), online shopping (31.6%), banking transactions (17.7%), and other applications (46%). According to the results of Chi-square test given in Table 6, MSc students use e-mail communication more that BSc students (20.3% more). Furthermore, gender has a statistically significant association with usage of Wi-Fi networks for banking transactions and other applications in terms that male students use it far more than female students.

In terms of security awareness, it is concerning that only half of participants stated that it is not safe to use Wi-Fi networks which require access password for sensitive activities (Q15). The big issue is that almost 41% of participants are not sure. Results of Chi-square test given in Table 6 show that there exists statistically significant association between gender and current education level and answer given on this question. Male and MSc students are the ones that mostly answered correctly.

Even greater issue is the knowledge provided by the participants in terms of transmission and browsing. Namely, two thirds of participants stated that they are not sure (59.3%) whether all Wi-Fi traffic is encrypted by default on all wireless routers (Q16) and that it is (24.4%). Only 16.3% answered correctly, that it is not encrypted. Among those that answered correctly, over 20% more are MSc students, while there was no difference in terms of gender. Also, 48.1% of participants are not sure regarding the meaning of https (Q17). Only 24.4% of them answered correctly, i.e., that the entered information is encrypted. Over 30% more correct

Table 6. Chi-square test results for transmission and browsing dimension (D2) questions related to cyber hygiene knowledge, awareness, and behavioral practices.

	Gender				Current Education Level			
	X-value	p-value	% Male	% Female	X-value	p-value	% BSc	% MSc
Q13.1 (B)	0.199	0.656	5.8	6.8	0.044	0.834	6.2	6.7
Q13.2 (B)	0.233	0.630	48.1	50.2	0.209	0.647	49.6	47.1
Q13.3 (B)	3.816	0.051	41.9	33.5	2.689	0.101	39.5	30.8
Q13.4 (B)	1.902	0.168	57.4	63.3	2.671	0.102	58.5	67.3
Q14.1 (B)	3.306	0.112	94.2	98.4	1.191	0.275	95.8	98.1
Q14.2 (B)	3.554	0.059	59.7	67.7	**14.742**	**< 0.001**	**59.5**	**79.8**
Q14.3 (B)	5.827	0.054	30.7	31.1	3.872	0.144	20.1	20.8
Q14.4 (B)	**8.278**	**0.004**	**22.5**	**12.7**	0.474	0.491	18.3	15.4
Q14.5 (B)	**23.742**	**< 0.001**	**56.6**	**35.1**	0.002	0.967	45.9	46.2
Q15 (A)	**4.768**	**0.029**	**53.1**	**43.4**	**4.588**	**0.032**	**45.9**	**57.7**
Q16 (K)	<0.001	0.986	16.3	16.3	**28.821**	**<0.001**	**11.9**	**33.7**
Q17 (K)	**9.787**	**0.002**	**30.2**	**62.9**	**18.211**	**< 0.001**	**20.2**	**40.2**
Q18 (K)	3.334	0.068	52.7	44.6	0.262	0.609	48.1	51.0
Q23 (K)	3.486	0.178	18.7	13.5	0.955	0.620	16.8	13.5
Q26 (K)	0.546	0.460	9.2	11.2	**4.116**	**0.042**	**11.6**	**4.8**

Abbreviations: A, Awareness; B, Behavior; BSc, Bachelor's degree; K, Knowledge; MSc, Master's degree; Q, Question.

answers were given by female students, and over 20% more by MSc students. This means that there is statistically significant association between gender and current education level and correct answer on question Q17. The remaining third rounded all other incorrect options.

Similarly concerning situation is related to their knowledge about botnets. Question 23 has been answered correctly only by 16.2%, while over 80% offered incorrect answer or was not sure when it comes to the question how the group of computers which is networked and used by hackers to steal information is called. Furthermore, only 10.1% of participants (mostly MSc students) correctly answered the question regarding the security risks reduced by using VPNs, which is that they reduce the use of insecure Wi-Fi networks (Q26). When asked if the “private search” option will protect them from Internet Service Providers (ISPs) seeing their activity, only half of participants answered that they will which is correct (Q18).

Next covered dimension is the one related to Facebook and social media (Q12). In terms of behavior on social media, most participants stated that they share name and surname and images. Half of them share videos and opinions, feelings, and statements. One third shares phone number and location, while less than 20% share other given information. There is no statistically significant association between gender and current education level and personal information sharing habits.

Authentication and credential dimension in terms of behavior is covered by Q9 and Q10. 50.8% of participants create their passwords by combining letters, numbers and symbols and use 8 characters. There is no statistically significant association between gender and current education level and password creation. 53.6% of them remembers their passwords, while 37.7% use the same passwords for most occasions and 25.3% store them on computer or write them down on paper which is quite concerning security behavior. Among those that remember the passwords, there is 19% more male students, and 36% more MSc students. This is in line with Chi-square test results given in Table 7 which show that there is statistically significant association between remembering passwords and gender and current education level. However, when it comes to passwords, the knowledge of participants is quite satisfying (regardless of gender and current education level) given that 84% of them correctly answered the question that tested what their opinion regarding the most secure password (Q11) was. That knowledge is not so good regarding recognition of two-factor authentication. Q22 question has been answered correctly by only 15.3%, one third was not sure, while the remaining participants answered incorrectly. Those that answered correctly were more male students (8% more). The same situation is with the knowledge related to ransomware, since Q24 question has been answered correctly by only 17.3%, 58.6% of them were not sure, while the remaining participants answered incorrectly. Those that answered correctly were more male students (17% more).

Finally, e-mail and messaging knowledge is not so well also. Namely, when asked if all e-mails are encrypted by default, only 7.5% of participants answered correctly (Q19). Similar situation is with the knowledge of phishing, since Q21 question, which addressed that notion, was correctly answered only by 12.8% of participants. After

Table 7. Chi-square test results for authentication and credential dimension (D4) questions related to cyber hygiene knowledge, awareness, and behavioral practices.

	Gender				Current Education Level			
	X-value	p-value	% Male	% Female	X-value	p-value	% BSc	% MSc
Q9 (B)	4.499	0.480	-	-	9.212	0.101	-	-
Q10.1 (B)	**17.266**	**< 0.001**	**17.4**	**33.5**	0.446	0.504	24.7	27.9
Q10.2 (B)	3.564	0.059	33.7	41.8	**11.222**	**0.001**	**34.1**	**51.9**
Q10.3 (B)	1.906	0.167	7.8	4.8	0.044	0.834	6.2	6.7
Q10.4 (B)	0.953	0.329	1.2	0.4	0.052	0.820	0.7	1.0
Q10.5 (B)	**22.402**	**< 0.001**	**64.0**	**43.0**	**43.096**	**< 0.001**	**61.0**	**25.0**
Q10.6 (B)	2.128	0.345	5.0	3.2	1.645	0.439	4.7	1.9
Q11 (K)	0.687	0.407	82.6	85.3	0.510	0.822	83.7	84.6
Q22 (K)	**6.664**	**0.010**	**19.4**	**11.2**	0.079	0.778	15.5	14.4
Q24 (K)	**25.780**	**< 0.001**	**25.9**	**8.8**	0.350	0.080	18.8	11.5

Abbreviations: A, Awareness; B, Behavior; BSc, Bachelor degree; K, Knowledge; MSc, Master degree; Q, Question.

conducting Chi-square test to examine the association between gender and current education level and correct answers on Q19 and Q21 (e-mail and messaging dimension – D5), there was no statistically significant association (gender – Q19: X-value = 0.999, p-value = 0.318; gender – Q21: X-value = 0.465, p-value = 0.495; education level – Q19: X-value = 0.797, p-value = 0.372; education level – Q21: X-value = 1.274, p-value = 0.259).

It is important to explain the reporting of percentage results given in Tables 5–7, since they differ from the ones given in Table 4. Percentages in Table 4 are extracted from the whole survey sample (509 participants). Percentages given in Tables 5–7 are calculated by extracting the number of participants that gave the correct answers (e.g., Q15) or selected an option (Q14) from the whole sample. For example, Q15 was answered correctly by 48.3% participants and the reported percentages of male and female students in Table 6 have been calculated by extracting the number of male and female students that have answered correctly from the total number of male and female student participants. Items where multiple answers were enabled are marked by adding numbers after the main question number (e.g., Q14(1), Q14(2), etc.). Also, for multiple choice items without correct answer, such as Q7, Q9, Q27-30, we did not report percentages where there was no statistically significant association.

Answers to the first three research questions dealing with the level of cyber hygiene knowledge, awareness, and behavioral practices, are as follows. The part participants mostly have **acceptable cybersecurity behavior** (in terms of sharing the access to their devices, personal data, and usage of unsecure Wi-Fi networks), except their management of passwords which needs to be revised. However, their **awareness** is **not on the satisfactory level**. This is mainly because to the fact that they are not aware of dangers related to data backups and usage of unverified links, as well as usage of Wi-Fi networks for sensitive transactions. The light spot in this domain is the awareness of participants regarding downloaded files, as well as usage of security software on their devices and software updates. Finally, their **knowledge** regarding cybersecurity terms such as https, VPNs, password security, ransomware, two-factor authentication, botnets, etc. can be qualified as **very low**. The correct answers have been given by very low number of participants (7.5% minimum (Q19) to 24.4% maximum (Q17)). Also, it is important to notice that very large number of participants answered the questions with "not sure." Only exception in terms of knowledge is related to secure passwords (Q11), where 84% of participants answered correctly.

When it comes to examination of **relations between gender and current education level and cyber hygiene knowledge, awareness, and behavioral practices, one can conclude that the relations exist in some cases**. Namely, in terms of gender, there exists an association in one third of considered cyber hygiene knowledge and behavioral practices items and one quarter of considered cyber hygiene awareness items. In terms of current education level, the situation is similar, as given in Table 11. To conclude, it can be stated that there is a potential that cyber hygiene knowledge, awareness, and behavior is affected by gender and current education level of university students in Bosnia and Herzegovina, but it needs to be additionally tested in a focused study.

Also, there exists statistically significant relation between cyber hygiene behavior and awareness in some cases, so we characterize the relation as partial. To have clear conclusion, this relation

Table 8. Chi-square test results for relation between cyber hygiene behavioral practices and awareness.

	Computer Sharing (D1-Q4)		Smartphone Sharing (D1-Q5)		Use of Wi-Fi (D2-Q13)		Reason for Wi-Fi Usage (D2-Q14)	
	X-value	p-value	X-value	p-value	X-value	p-value	X-value	p-value
D1-Q7	**22.455**	**< 0.001**	4.556	0.336	-	-	-	-
D2-Q15	2.055	0.152	0.544	0.461	**11.095**	**0.001**	**5.081**	**0.024**
D1-Q20	2.911	0.573	6.266	0.280	-	-	-	-
D1-Q27	2.233	0.693	6.363	0.174	-	-	-	-
D1-Q28	4.685	0.321	**10.761**	**0.029**	-	-	-	-
D1-Q29	2.482	0.289	2.046	0.359	-	-	-	-
D1-Q30	**5.974**	**0.049**	1.190	0.551	-	-	-	-

Abbreviations: D, Dimension; Q Question.

needs to be examined in more detail in specifically focused study. To obtain answer to this research question we have grouped and analyzed the items referring to the same topic, while some of items were not considered since their type was not adequate to test the association. For example, we did not considered items regarding devices used for Internet usage since most participants use their desktop/laptop computers or smartphones (Q3). Then, data collected under items Q6 and Q12 is not adequate to test the relation to cyber hygiene awareness. Data collected under items Q9 and Q10 was not considered since we did not have awareness password questions. In summary, we tested the associations between computer (Q4) and smartphone (Q5) sharing habits and items Q7, Q15, Q20, and Q27-30. Also, we have tested the relation between Wi-Fi behavior items (Q13 and Q14) and awareness item regarding Wi-Fi usage (Q15). The results of Chi-square test are given in Table 8. The results show the statistically significant associations between computer sharing behavioral practices and awareness to enable security software on those devices and to use safe links on smartphones. Smartphone usage habits have statistically significant relation to awareness of smartphone software updating. Finally, Wi-Fi usage habits have statistically significant association with awareness on using Wi-Fi for mobile banking.

Similar conclusion can be made regarding **the relation between cyber hygiene behavioral practices and knowledge**. This **relation exists in some cases**, but it needs to be examined in more detail in a focused study. To reach these results we also used Chi-square test (Table 9) and tested some of the items. Data collected under items Q3, Q6, and Q12 has not been analyzed due to the same reasons stated above. We have analyzed the existence of relations between computer (Q4) and smartphone (Q5) sharing habits and cyber hygiene knowledge items (Q11, Q16-Q19, Q21-26). Then, we have analyzed the relation between password creation and management behavior (Q9 and Q10) and relevant knowledge data (Q11), and Wi-Fi usage data (Q13 and Q14) and relevant knowledge data (Q16). Computer and smartphone sharing is associated with cyber hygiene knowledge is some segments, i.e., knowledge regarding https, ransomware, and GPS. Password creation is associated with the correct answer regarding passwords, while password management is not. Finally, Wi-Fi usage habits are not associated with correct answer regarding Wi-Fi encryption.

Similarly, after the analysis conducted by using Chi-square test with **cyber hygiene awareness and knowledge** items, one can conclude that the results indicate the **partial existence of the relation**. Analyzed items from the awareness group are Q7, Q15, Q20, and Q27-30, while the items on the knowledge side are Q11, Q16-Q19, Q21-26. The results given in Table 10 show that awareness regarding enabling security solutions and data backup does not affect cyber hygiene knowledge since we found no statistically significant relation. There are statistically significant associations between awareness on Wi-Fi safety and correct answers on Q16-19 and Q23-24. Also, there are statistically significant associations between awareness on device and file checking and correct answer on Q11 and Q22. Software update awareness has

Table 9. Chi-square test results for relation between cyber hygiene behavioral practices and knowledge.

	Computer Sharing (D1-Q4)		Smartphone Sharing (D1-Q5)		Password Creation (D4-Q9)		Password Management (D4-Q10)	
	X-value	p-value	X-value	p-value	X-value	p-value	X-value	p-value
D4-Q11	1.641	0.200	0.522	0.470	**19.477**	**0.002**	1.678	0.567
D2-Q16	0.036	0.850	3.781	0.052	-	-	-	-
D2-Q17	**4.762**	**0.029**	0.284	0.594	-	-	-	-
D2-Q18	1.179	0.278	0.264	0.608	-	-	-	-
D5-Q19	3.629	0.057	1.358	0.244	-	-	-	-
D5-Q21	0.139	0.709	0.002	0.946	-	-	-	-
D4-Q22	0.482	0.488	0.317	0.573	-	-	-	-
D2-Q23	2.290	0.318	2.392	0.302	-	-	-	-
D4-Q24	**6.598**	**0.010**	**4.361**	**0.037**	-	-	-	-
D1-Q25	**6.554**	**0.038**	3.772	0.152	-	-	-	-
D2-Q26	1.188	0.276	2.605	0.107	-	-	-	-

Abbreviations: D, Dimension; Q, Question.

statistically significant association with Q11 and Q19 correct answers. Finally, there is a statistically significant association between awareness regarding the usage of safe links on computers and smartphones and the correct answers to the following questions Q11, Q19-22, Q26 and Q11, Q16, Q21-24, and Q26, respectively.

5. Discussion

In this section, we have compared the results of our study with the existing literature and described the limitations of the study. The concluding comments along with the recommendations resulting from them are given in the conclusion section.

Our study is based on the literature related to five dimensions (i.e., storage and device, transmission and browsing, Facebook and social media, authentication and credentials, and e-mail and messaging) and three outcomes (i.e., behavior, awareness, knowledge) of cyber hygiene. To our knowledge, this is the first study to investigate aforementioned dimensions and outcomes of cyber hygiene in Bosnia and Herzegovina. Research results show that university students in Bosnia and Herzegovina have acceptable cyber hygiene behavior. However, their awareness of cyber hygiene is not satisfactory, as well as their knowledge, which is quite low.

The research results show accepted behavioral practices of university students in Bosnia and Herzegovina, especially in storage and device hygiene (D1), as well as in transmission and browsing hygiene (D2). However, Facebook and social-media hygiene (D3) requires alternations in behavioral practices, since uploading of great amount of personal information is identified as a main privacy issue (Debatin et al., 2009). Furthermore, user behavior in terms of authentication and credential hygiene (D4) can be revised, especially password usage and management. This result is in line with previous study given by Grawemeyer and Johnson (2011), which found that lax security behavior included password reuse, lack of knowledge about the (in)secure passwords, sharing or writing down the passwords. The reason for such behavior was identified by Whitty et al. (2015) who have found factors that predict the risky practice of sharing passwords, i.e., age, perseverance, and self-monitoring. Younger people share passwords more often than older people (Whitty et al., 2015). Reason probably lies in fact that they have more family and friends who are active online compared with older people. Therefore, the educational campaigns need to focus on young people as their target audience. Moreover, they should focus on a willingness among users to improve their knowledge on password-related matter and apply these secure password practices (Butler, 2014). Similar, this behavior in terms of passwords is reported by Olmstead and Smith (2017), where 75% of survey participants can identify the most secure password from the list of several options.

Our findings show that university students in Bosnia and Herzegovina had an unsatisfactory awareness level of cyber hygiene. We found that relatively small percentage of participants perform basic practices related to storage and device hygiene (D1) (e.g., installing antivirus or antimalware software on various devices, regularly performing data backups, etc.) and transmission and browsing hygiene (D2) (e.g., use safe and checked links and network). It is a devastating fact given that it is expected to be worse with other people without university background. In this context, the most smartphone users trust the official repository when downloading applications, therefore, security controls are not enabled or not added, and users disregard security during application selection and installation (Mylonas et al., 2013). As with previous study given by Alani (2017), low awareness of cybersecurity makes smartphone users vulnerable to different attacks. However, Zwilling et al. (2020) found that even partial attendance in cybersecurity program has positive impact on level of awareness of cyber hygiene. This highlights the importance of educational trainings and programs in cyber hygiene. In this regard, Dodge et al. (2007) showed that phishing exercises raised security awareness, whereas Arachchilage and Love (2014) identified the reason why computer users aged between 18 and 25 more were vulnerable to phishing threats (i.e., lack of knowledge).

The research results indicate that university students in Bosnia and Herzegovina have low level of knowledge regarding cyber hygiene. In terms of storage and device hygiene (D1), less than half of participants provide correct answers

Table 10. Chi-square test results for relation between cyber hygiene awareness and knowledge.

	Security Solutions Enabled (D1-Q7)		Wi-Fi Safety (D2-Q15)		Device and File Checking (D1-Q20)		-	
	X-value	p-value	X-value	p-value	X-value	p-value	-	-
D4-Q11	4.127	0.389	0.155	0.694	**9.740**	**0.045**	-	-
D2-Q16	1.275	0.866	**18.442**	**< 0.001**	3.076	0.545	-	-
D2-Q17	4.184	0.382	**22.725**	**< 0.001**	1.174	0.882	-	-
D2-Q18	9.231	0.056	**12.775**	**< 0.001**	6.799	0.147	-	-
D5-Q19	1.422	0.840	**15.247**	**< 0.001**	8.496	0.075	-	-
D5-Q21	4.636	0.327	0.016	0.901	6.259	0.181	-	-
D4-Q22	2.591	0.629	3.591	0.058	**11.948**	**0.011**	-	-
D2-Q23	6.222	0.622	**6.409**	**0.041**	6.829	0.555	-	-
D4-Q24	7.020	0.135	**5.251**	**0.022**	7.145	0.128	-	-
D1-Q25	4.065	0.851	2.673	0.263	9.106	0.333	-	-
D2-Q26	1.403	0.844	0.280	0.597	5.629	0.229	-	-
Data backup (D1-Q27)			Software update (D1-Q28)		Safe links on computers (D1-Q29)		Safe links on smartphones (D1-Q30)	
	X-value	p-value	X-value	p-value	X-value	p-value	X-value	p-value
D4-Q11	5.155	0.272	**12.062**	**0.017**	**7.566**	**0.023**	**15.520**	**<0.001**
D2-Q16	5.629	0.229	6.459	0.167	4.016	0.123	**7.044**	**0.030**
D2-Q17	9.005	0.061	8.967	0.062	0.442	0.802	2.405	0.300
D2-Q18	5.297	0.258	2.971	0.563	0.816	0.665	5.892	0.053
D5-Q19	8.883	0.064	**16.624**	**0.002**	**14.681**	**0.001**	2.617	0.270
D5-Q21	3.780	0.437	4.138	0.388	**7.572**	**0.023**	**6.738**	**0.034**
D4-Q22	4.607	0.330	0.712	0.950	**6.709**	**0.035**	**7.802**	**0.020**
D2-Q23	12.866	0.117	1.217	0.996	6.284	0.179	**9.859**	**0.043**
D4-Q24	1.416	0.841	5.350	0.253	5.902	0.052	**9.987**	**0.007**
D1-Q25	8.237	0.411	15.011	0.059	2.686	0.612	5.839	0.212
D2-Q26	1.357	0.852	7.015	0.135	**16.149**	**< 0.001**	**6.502**	**0.029**

Abbreviations: D, Dimension; Q, Question.

regarding positioning and tracking possibilities, whereas even greater issue is the knowledge provided in terms of transmission and browsing hygiene (D2). As with previous study given by Imgraben et al., 2014) participants have poor knowledge of the risks associated with leaving Wi-Fi turned on the whole time. Younger generation would be the most likely to connect to an unknown or unsecure Wi-Fi network. According to Markelj and Bernik (2015), the use of security solutions derives from knowing the threats. Our findings show that participants are familiar with general threats, whereas they have poor knowledge of complex threats (e.g., botnet, ransomware, etc.). The same is true for the e-mail and messaging hygiene (D5), as well as authentication and credential hygiene (D4). This is in line with previous study given by Florêncio et al. (2007), who found that strong passwords did not protect the online users from password stealing threats, such as keylogging and phishing. The results obtained in this study are in line with the results reported by Olmstead and Smith (2017). For example, Olmstead and Smith (2017) report that only 16% of participants know that a botnet is a networked set of computers used for criminal purposes, or only 13% recognizes that a VPN minimizes the risk of using insecure Wi-Fi.

Our study also shows that students from Bosnia and Herzegovina have the similar cyber hygiene knowledge as general population of Americans, since Olmstead and Smith (2017), which used a part of questions answered in this study (13 questions), report that many Americans are unclear about some key cybersecurity topics, terms, and concepts. Most of online adults in USA can identify a strong password when they see one and recognize the dangers of using public Wi-Fi. However, many struggle with more technical cybersecurity concepts, such as how to identify true two-factor authentication or determine if a webpage they are using is encrypted. Also, Olmstead and Smith (2017) state that knowledge on cybersecurity varies by several demographic factors. For example, those with college degree offer more correct answers. Cybersecurity knowledge also varies by respondent's age, although these differences are much less dramatic than the differences pertaining to education attainment.

When it comes to results regarding the relation between gender and cyber hygiene outcomes in Bosnia and Herzegovina, our results are in line with previously conducted studies with students or employees (Anwar et al., 2016; Fatokun et al., 2019; Kshetri & Chhetri, 2022; Peacock & Irons, 2017). In these studies, as well as in our, gender showed to have the potential to be important in terms of cyber hygiene behavior, awareness, skills, habits, experiences, etc. Namely, the results of our study show that male students in higher percentages select correct answers to cyber hygiene knowledge questions, except the one related to https, but female students have more awareness and better behavior. As suggested by Anwar et al. (2016) that examined gender in terms of employees, this could lead to gender-specific approach to cyber hygiene matters and models. Also, it can be seen as an opportunity for women, as suggested by Peacock & Irons, 2017), since cybersecurity is mainly perceived as "man's job." However, an opposite conclusion has been made in comparison to the existing literature when it comes to current education level effects on cyber hygiene outcomes. For example, Fatokun et al. (2019) finds that Malaysian students' cyber hygiene is not affected by their education level to a large extent, while our study shows some cases in which the relation exists (Table 11). This calls for awareness in the institutions of all education levels, but enterprises and organizations as well, to train and update the knowledge of students and employees with the common and state-of-the-art cyber hygiene knowledge, awareness, and behavioral practices. Our results are in some cases closer to Olmstead and Smith (2017) research results in terms that MSc students give more correct answers than BSc students. However, this needs to be taken

Table 11. Summary of answers to research question.

Research Question	Answer
What is the level of cyber hygiene knowledge?	Very low
What is the level of cyber hygiene awareness?	Not satisfactory
What is the level of cyber hygiene behavioral practices?	Acceptable
Is there a relation between gender and cyber hygiene knowledge, awareness, and behavioral practices?	Relation exists in some cases: • Cyber hygiene behavior (3 cases out of 9): There is a relation in sharing smartphone devices and applications that are used on smartphones, Wi-Fi usage habits, and password management. • Cyber hygiene awareness (2 cases out of 8): There is a relation in scanning files and devices and Wi-Fi usage safety. • Cyber hygiene knowledge (4 cases out of 11): There is a relation in knowledge regarding https, GPS, two-factor authentication, and ransomware.
Is there a relation between current education level and cyber hygiene knowledge, awareness, and behavioral practices?	Relation exists in some cases: • Cyber hygiene behavior (3 cases out of 9): There is a relation in sharing computer and smartphone devices, Wi-Fi usage habit, and password management. • Cyber hygiene awareness (2 cases out of 8): There is a relation in scanning files and devices and Wi-Fi usage safety. • Cyber hygiene knowledge (3 cases out of 11): There is a relation in knowledge regarding https, Wi-Fi encryption, and VPNs.
Is there a relation between cyber hygiene behavioral practices and awareness?	Relation exists in some cases. There is a relation between: • Computer sharing behavior and awareness to enable security software on those devices and to use safe links on smartphones. • Smartphone usage habits and awareness of smartphone software updating. • Wi-Fi usage habits and awareness on using Wi-Fi for mobile banking.
Is there a relation between cyber hygiene behavioral practices and knowledge?	Relation exists in some cases. There is a relation between: • Computer and smartphone sharing, and cyber hygiene knowledge is some segments (knowledge regarding https, ransomware, and GPS); • Password creation and the correct answer regarding passwords.
Is there a relation between cyber hygiene knowledge and awareness?	Relation exists in some cases. There is a relation between: • Awareness on Wi-Fi safety and knowledge on Wi-Fi and e-mail encryption, https, private search, botnet, and ransomware. • Awareness on device and file checking and knowledge regarding secure password and two-factor authentication. • Software update awareness and knowledge on secure passwords and e-mail encryption. • Awareness regarding the usage of safe links on computers and smartphones and most cyber hygiene knowledge questions (except ones related to https and GPS).

Abbreviations: GPS, Global Positioning System, VPN, Virtual Private Network, Wi-Fi, Wireless Fidelity.

with caution given that Olmstead and Smith (2017) addressed broader population than university students.

Our results related to the existence of relations between cyber hygiene behavioral practices, awareness, and knowledge in some cases (tested on Bosnia and Herzegovina's university students) are in line with the findings given by Zwilling et al. (2020) who tested participants in Israel, Slovenia, Poland, and Turkey (different gross domestic product (GDP) values). Both studies state that there is an association between cyber hygiene awareness and knowledge, behavior and knowledge, and behavior and awareness. Additional value of this finding is that these relations are existing regardless of GDP and cultures. The existing relations reported in the results section are mainly, but not necessary, between the items from the same cyber hygiene dimension group. However, it is important to note that in this case, the acceptable cyber hygiene behavior does not result in acceptable level of awareness and knowledge, but low awareness is associated to low knowledge and vice versa.

Cybersecurity is not solely a technical issue, but also complex of human-related issues which consequently influences our quality of life and quality of experience toward (Baraković et al., 2020). If we have a positive perception of cybersecurity when we use the technology, we have greater chance to have more quality life in digital dominant society. Also, it is important to mention that human beliefs are associated with the extent to which an individual absorbs knowledge (David et al., 2020). The requirement to improve the cyber hygiene of average individual is of equal importance as the requirement for professional programs focusing on education and training in the cybersecurity field (Dupuis, 2017). These programs can be provided in an engaging format, such as flexible and highly interactive video game to be an effective addition to basic information training programs (Cone et al., 2007). Finally, targeted gender-dependent and education-independent trainings and programs should be provided to all to maintain knowledge of the last cybercrime activities and the best cyber protection measures available. They will be more effective if they provide an understanding why this is important (Parsons et al., 2014).

This study has some limitations that should be addressed in future research. One limitation is that findings may be affected by the participant's demographics. There is a possibility that significant differences exist in the cyber hygiene outcomes of participants between different demographics. For example, all participants were young age and tend to adopt technology, which may influence the study analysis. This leads to an open research item for future research, i.e., conducting this survey among different age and education groups to gain clearer picture of situation regarding cybersecurity in Bosnia and Herzegovina.

Additionally, data collection was based on self-reported statistics (i.e., participants were asked to complete the survey). Although a direct validation of the collected responses was avoided, the discussion with the experts and participants provided the understanding of the questions and the responses' validity. Participants were not informed that the final purpose of the survey was to evaluate the cyber hygiene outcomes (i.e., behavior, awareness, knowledge), since it is expected that this would pump-up their responses. Therefore, in the future, experiment studies testing the cyber hygiene knowledge, awareness, and behavioral practices of students, but public as well, should be conducted.

Finally, to our best knowledge, this is the first study focusing on the evaluation of cyber hygiene outcomes (i.e., behavior, awareness, knowledge) together with other impacts and interplays among university students in Bosnia and Herzegovina. Therefore, future studies should further explore our findings referring to the unsatisfactory level of awareness and knowledge regarding cyber hygiene and possible relationships between different influential factors and cyber hygiene outcomes (as well as their interplay).

6. Conclusion

In an increasingly digital world and COVID-19 pandemic world, where there is an increase in cybersecurity incidents and threats, personal data can be as valuable and as vulnerable to individuals who have wrong intentions as any other possessions. This can be mitigated by good cyber hygiene habits. The main aim of this paper was to evaluate, analyze, and understand the level of knowledge, awareness, and behavioral practices on cyber hygiene of a group of citizens, i.e., students in Bosnia and Herzegovina to have the clearer picture on cyber hygiene in Bosnia and Herzegovina. Also, we aimed to examine the relation among cyber hygiene knowledge, awareness, and behavior, and find out do gender and current education level affect them.

For those purposes, we have conducted a survey study containing 30 cyber hygiene related questions. Our study is based on literature related to five dimensions (i.e., storage and device, transmission and browsing, Facebook and social media, authentication and credentials, and e-mail and messaging) and three outcomes (i.e., behavior, awareness, knowledge) of cyber hygiene. To our knowledge, this is the first study to investigate aforementioned dimensions and outcomes of cyber hygiene in Bosnia and Herzegovina. Research results show that university students in Bosnia and Herzegovina have relatively acceptable cyber hygiene behavior, but their awareness of cyber hygiene is not satisfactory, as well as their knowledge, which is quite low. Also, the study shows the existence of some relations between gender and current education level and cyber hygiene knowledge, awareness, and behavior, as well as the mutual interplay and relations between the cyber hygiene outcomes.

Contributions of this paper are bi-directional: theoretical and practical. Theoretical contributions of the paper reflect in recognized level of cyber hygiene behavior, awareness, and knowledge among university students in Bosnia and Herzegovina and determined need to support the cyber hygiene practices and alteration of perception of cybersecurity challenges. Also, from the theoretical point of view, the results of this study can serve as a starting point for the discussion, policies, and strategies on cybersecurity in Bosnia and Herzegovina and more intense work in improving the cyber hygiene. However, given that the results of the study do not differ from the ones reported in studies in other countries, it can initiate the broader discussion and engagement in the field of cyber hygiene in the Western Balkan and European region, as well as worldwide. Consequently, the results of this study contribute to EU integration path of Bosnia and Herzegovina.

Furthermore, the results regarding the existence of relations between gender and current education level and cyber hygiene knowledge, awareness, and behavior, as well as mutual interplay and relations between cyber hygiene outcomes, can shape the future of cyber hygiene and serve as a basis for future focused research studies aimed at improving the level of cybersecurity from many aspects.

On the other hand, practical implications of this study are to apply them for development of cyber hygiene program in Bosnia and Herzegovina and provide guidelines both for organizations and individuals to protect themselves from cyber threats. In this term, they focus on proactive measures to improve user experience and security, such as: (i) limit user access to hardware/software, (ii) update hardware/software regularly, (iii) use antivirus/antimalware tools, (iv) manage complex passwords, (v) use multi-factor authentication, (vi) backup regularly, (vii) avoid suspicious sites and e-mail attachments, (viii) make cyber hygiene part of routine. Furthermore, practical implications refer to development of specific educational trainings and programs with the purpose to: (i) increase the knowledge regarding cyber hygiene concepts and issues, (ii) develop new attitudes toward cybersecurity threats, (iii) convert cybersecurity awareness into behavioral practices, (iv) build new rules sharing good practices for cyber hygiene. In addition, cyber hygiene programs need to be gender-aware to provide the opportunity for both sexes to gain the same knowledge and awareness levels in terms of cybersecurity.

Open research items which stem from this study include the validation of the results of this study, but also the validation of the survey. To get clearer picture on the level of cyber hygiene in Bosnia and Herzegovina, but worldwide as well, future research should conduct additional surveys covering the rest of population (different age, profession, and education groups) and explore our findings referring to the unsatisfactory level of awareness and knowledge regarding cyber hygiene. Also, as stated earlier, focused studies should be conducted to additionally test and verify the relations between various factors (gender, education level, age, culture, etc.) and cyber hygiene outcomes, as well as mutual relations and dependencies among segments or cyber hygiene knowledge, awareness, and behavior in total. This study should be exploratory in nature. Finally, this study should initiate development programs for improving cyber hygiene through education and trainings which can be used worldwide, since cyber space is borderless and will be increasingly used in this, i.e., COVID-19 or coming pandemic circumstances. Furthermore, maintaining security and protection in digital world is our common challenge.

Disclosure statement

We declare that the submission contains 100% original work which has not been published previously and that it is not under consideration for publication elsewhere. We wish to confirm there are no known conflicts of interest associated with this study that could inappropriately influence, or be perceived to influence, this work. We further confirm that the manuscript has been read and approved by both named authors and that there are no other persons who satisfied the criteria for authorship but are not listed. We further confirm that the order of authors has been approved by both of us.

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